2023 5th International Conference on Bio-engineering for Smart Technologies (BioSMART) | 979-8-3503-3849-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/BioSMART58455.2023.10162074

HViT4Lung: Hybrid Vision Transformers Augmented by Transfer Learning to Enhance Lung Cancer Diagnosis

Reza Roofegari nejad
Aspen Technology Inc., Bedford, MA, USA
r.roofegari@knights.ucf.edu

Sahar Hooshmand
Department of Computer Science
California State University-Dominguez Hills, Carson, CA, USA
hooshmand@csudh.edu

***Abstract*—Lung cancer is the leading cause of mortality among other forms of cancer worldwide. Early and accurate recognition of lung nodules is crucial for the better life quality of patients. Although the chest Computed Tomography (CT) scan is the principal imaging procedure to evaluate and recognize lung cancer, the radiologists evaluation based on CT images is subjective and afflicted from a low accuracy compared to the post-surgery pathological tests. Computer Aided Diagnosis (CAD) has been proven to be beneficial in this context by increasing accuracy and minimizing expert involvement. Nevertheless, due to various factors including size and location inconsistency of nodules, the errorless detection of cancerous cases is still a challenge for CAD systems. Motivated by this fact, this paper presents a novel and effective method, called HViT4Lung (Hybrid Vision Transformers for Lung cancer detection), to enhance lung cancer diagnosis. The proposed deep learning-based hybrid framework combines Transformers and Convolution Neural Networks, augmented by transfer learning that extracts features from the chest CT images to detect lung nodules and predict their malignancy. The proposed pipeline of HViT4Lung is implemented with various blocks of deep learning and tested on a sample lung CT images dataset. The results of the proposed model are very promising compared to the other approaches in this field, achieving 99.20% training accuracy, 99.09% validation accuracy, and 99.09% testing accuracy for the classification of lung CT scans of 1190 images into three different classes of normal, benign, and malignant.**

***Index Terms*—Vision Transformers, Convolution Neural Networks, Deep Learning, Image Classification, Computer-aided Diagnosis**

I. Introduction

World health organization reports lung cancer as one of the leading causes of death worldwide, among all other cancers. The American Cancer Society (ACS) statistics show 1,762,450 new cancer cases in 2019, leading to 606,880 deaths in the United States, where 13% of all these new cases are lung cancer [1]. Nevertheless, this rate is even higher in developing countries. One of the essential screening issues related to this kind of cancer is the difficulty in diagnosing the disease in its early stage. Reports and data show that timely diagnosis and early identification of this cancer can help in increasing the patient's chance of survival. In fact, the identification of lung cancer through Computed Tomography (CT) scan decreased the mortality rate by 20% and with the recent development in scanning equipment, the identification of this illness has increased significantly [2].

Various factors including observer error, decision-making fault, technical aspects error, complicated structure of the lung, and tumor characteristics such as lesion size and its location can contribute to a misdiagnosis of lung cancer [3]. Furthermore, some studies are utilizing thin slices (less than 1.5 mm) and high-resolution CT images [4], [5] while others are using low-dose scanning protocols with non-thin slices (less than 5 mm) [6]. Therefore, healthcare experts have recently emphasized on establishing an advanced method that performs well under different scenarios, despite the technical settings. Basically, the non-automatic analysis of lung abnormality derived from CT scan images could be an inaccurate and time-consuming procedure. A more efficient method would be to evaluate the lung deformities by computer-aided strategies and then submit the report to the specialist to improve the accuracy of diagnosis and reinforce the treatment steps.

Recent developments in computer vision and specifically deep learning methods lead to significant achievements in many fields of computing and medicine including medical imaging. This process is even accelerated by the introduction of powerful processors and GPUs which enables solving very complex problems in short time periods. Accordingly, introducing state-of-the-art deep learning techniques in the medical imaging field has equipped doctors with computer-aided diagnosis (CAD) that enhances the process, especially in the early stages of the illness which is difficult to detect.

The introduction of deep Convolutional Neural Networks (CNN) especially 3D-CNNs has shown considerable progress in image classification [7], object detection [8], and visual tasks. Moreover, utilizing Vision Transformers (ViT) [9] in various image processing problems, including medical image analysis causes a significant improvement in CAD for different types of diseases as well as lung cancer.

This paper presents a novel idea of using the Hybrid Vision Transformer model (HViT) with convolutional neural networks augmented by transfer learning methodology to analyze and classify lung CT scan images. The proposed approach which is called "*HViT4Lung*" is a feature-based classification method that combines the attention mechanism and global dependency of the input-output of transformers model, the feature detection power of CNNs, and knowledge sharing of transfer learning approach to classify the lung CT scan images accurately. The

Authorized licensed use limited to: University of Florida. Downloaded on October 30,2026 at 04:54:02 UTC from IEEE Xplore. Restrictions apply.

key contributions of this paper are:

1) proposing a pipeline by combining transformers and CNN models for improving computer-aided diagnosis
2) applying the proposed model on a real lung CT scan dataset and achieving a high classification accuracy

The rest of the paper is organized as follows. Section II reviews the related literature and work of this research study. Section III provides background and the methodology of the proposed HViT4Lung pipeline. Then, in Section IV, the dataset details, implementation architecture, and experimental results are discussed. Finally, Section V concludes the paper and highlights the future direction.

II. Related Work

Lung nodules can be tiny and seem identical to the structure of blood vessels or benign tumors [10]. Therefore, lung CT scan images have a high false-positive rate [11]. Recently, several Deep Learning-based methods have been developed to analyze lung CT scan images which improved the accuracy of tumor lesion detection significantly [12]. For instance, Sun et al. utilized CNNs, Stacked Denoising Autoencoder (SDAE), and Deep Belief Networks (DBNs) on Lung Image Database Consortium (LIDC) image collection and the accuracies were 79.7%, 79.2%, and 81.1% respectively [13]. Also, in 2018, lung nodules from Image Database Resource Initiative, and Lung Image Database Consortium, has been recognized with an accuracy of 97.6% [14]. Faisal et al. applied several classifiers, including Support Vector Machine, multi-layer perceptron, and decision tree on UCI repository dataset for predicting the early-stage of lung cancer [15]. In their study, ensemble models were utilized to compare the performance where the gradient-boosted tree achieved the best accuracy of 90%.

Later, Hofmanninger et al. proved that the result of deep learning based analysis of lung CT images can significantly be improved by training the models using a variety of datasets [16]. Their open-source lung segmentation AI model was useful to remove the false positives in a study of Covid-19 lung abnormalities [17]–[19]. As aforementioned, one vital step in accurately detecting lung nodules is to efficiently minimize the false-positive rate. Recent studies tackled this problem by using various techniques, including 3D-CNN [20], multi-checkpoint ensemble [21], multi-scale prediction [22], relation learning [23], and multi-scale attention [24].

As a consequence of detecting the lung nodules, several techniques were presented to further analyze them. Some of these methods predicted the malignancy of the tumors. For instance, in 2021, Vieria et. al reached an accuracy of 92% for predicting the presence of cancerous tumors by building a data mining classification model based on CNNs [25]. There were other successful studies to predict the malignancy of lung tumors including the Multi-crop CNNs introduced by Shen et al. [26] and Progressive Growing Channel Attentive Non-Local (ProCAN) network introduced by Al-Shabi et al. [27]. Additionally, some studies predicted the sub-types of the tumor malignancy, by introducing the multi-view multi-scale CNN model [28] and leveraging a classification method based on hybrid features [29].

Recently, Dosovitskiy et. al developed the vision transformers model which analyzes the non-overlapped patches of an image as a sequence, and generated state-of-the-art results in image classification [9]. Afterward, ViT has been successfully utilized in several image processing problems including the analysis of medical images [30], [31]. In 2022, Niu et al. leveraged the ViT model and unsupervised contrastive learning for detecting lung nodules on CT scan images. Their experimental results have indicated the superiority of their approach over the advanced 3D-CNN baselines [32].

In this paper, we present our accurate and novel HVit4Lung method to detect lung cancer based on CT images. Our experimental results demonstrate the achievement of 99% accuracy, which is very inspiring and has superior performance in comparison with other models in this field.

III. HViT4Lung for Lung Cancer Diagnosis

A. Background

This section explains the proposed transformer based pipeline as a feature-based classification method for detecting and diagnosing lung cancer. Transformers were originally proposed in [33] for natural language processing (NLP) applications and become the state-of-the-art method in sequence modeling and transduction problems such as machine translation. The method is inspired by the attention mechanism for recurrent neural networks (RNN) which allows the model to focus on a special part of the sequence input data. However, unlike RNN models, the transformers refrain the recurrence feature and draw global dependencies between input and output. This characteristic provides a significant potential for parallelization in processing recurrence data similar to the CNN models.

Transformers are based upon the self-attention mechanism to capture dependencies among various instances of the input sequence data [33]. The self-attention is based on the scaled dot-product of the attention function which maps a query q_i of input token i and a set of keys k_j to the output. Then, the final output is calculated based on the weighted average of values v_i and the attention function. Each input token is multiplied by three learnable weight matrices to form related queries, keys, and values. The attention calculation for all tokens of the input sequence can be expressed as one large matrix calculation [33] using the softmax function as:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V \qquad (1)$$

where the ith rows of matrices Q, K, and V are represented as q_i, k_i, and v_i, respectively. Also, d_k is the dimension of the input queries and a key that prevents the speedy growth of the attention function by large inputs to maintain computational stability.

It was found beneficial to linearly project the queries, keys, and values, h times with different learnable matrices to input sequence with dimensions d_k, d_k, and d_v [33]. In fact, each

Authorized licensed use limited to: University of Florida. Downloaded on October 30,2026 at 04:54:02 UTC from IEEE Xplore. Restrictions apply.

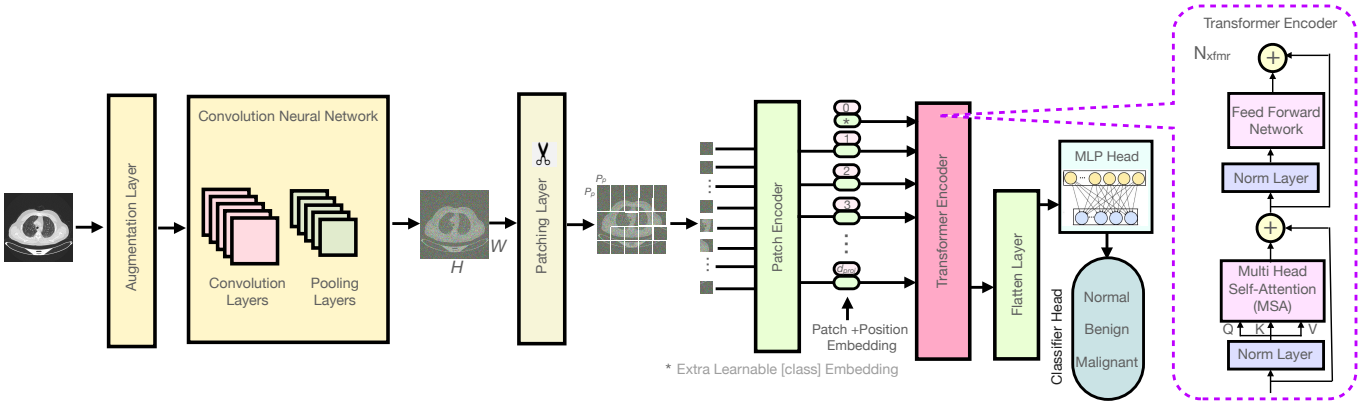


Fig. 1. Pipeline of the proposed HViT4Lung

set of these learnable matrices is called a head while the transformer model has a Multi-head Self-Attention (MSA) layer. These output heads are concatenated as equation (2) to form a single output and pass it into the feed-forward layer. The feed-forward layer is a multi-layer perception (MLP) network that helps to detect various features within the input sequence. Such structure provides significant performance and potential of the parallel processes for sequence data, especially in NLP applications.

$$MSA(Q, K, V) = Concat(head_1, ..., head_h)W^O \qquad (2)$$

Recently, the transformer model attains significant attention and shows superior performance by directly applying to image processing applications especially classification approaches [9]. These models are called Vision Transformers (ViT) and are realized by splitting an input image into patches and feeding the sequence of linear embedding of these patches into the model. In fact, these patches play a token role in the sequence data applications of the transformer model.

B. Proposed HViT4Lung Pipeline

The structure of the proposed ViT model adheres to the architecture of the NLP transformers to use their scalability and efficient implementation. Fig. 1 shows the pipeline of the proposed hybrid HViT4Lung which combines CNN and ViT networks. The pipeline is enhanced by forming the ViT input sequence patches from feature maps of a CNN layer. In this pipeline, a CNN network similar to the LeNet-5 [34] consisting of multiple convolution and average-pooling layers is used. It is worth mentioning that a data augmentation layer is added before the CNN layer to bold specific features and help prevent any possible overfitting of the model on the train set. This layer which is shown in the pipeline structure consists of normalization and resizing layers for input data as well as augmenting data by random flipping, rotating, and zooming.

According to the pipeline, the output of the CNN layer is fed into the patching layer. The standard transformer receives a single-dimensional sequence of token embeddings. To adapt this model with images of $\mathbb{R}^{H\times W\times C}$ dimensions, the patching layer reshaped it into a sequence of flattened patches of size $\mathbb{R}^{N_p({P_p}^2.C)}$. Here, (H, W) is the resolution of the input image, C is the number of channels, and $(P_p \times P_p)$ is the resolution of each patch. Therefore, the number of patches can be achieved by (3). It is worth mentioning that these dimensions have an analogy to the original transformer model which is described in previous subsections. Also, the dimensions of the picture and patches should be carefully set to have a positive integer number of patches.

$$N_p = \frac{H \times W}{{P_p}^2}, \qquad N_p \in \mathbb{Z}^+ \qquad (3)$$

Considering the pipeline, the patches are fed into the patch encoder layer. This layer flattened the patches and linearly transforms them by projecting into a vector of size d_{proj}. Also, similar to the sequence data, this layer adds a learnable positional encoding to the projected vector in order to retain the positional information of the patches. The resulting sequence of embedding vectors serves as an input for the transformer encoder layer.

The transformer encoder layer consists of multiple N_{xfmr} transformer blocks. According to the pipeline figure, each block includes MSA Layer, the feed-forward MLP layer, and the norm layer [9]. The norm layer along with a residual connection is applied before each MSA and feed-forward layers which helps to speed up the learning process. This transformer encoder layer is similar to the original transformer model which is described in the previous subsection. The output of the transformer encoder block is a vector with the dimension of $(batch_{size}, N_p, d_{proj})$ which serves as an input for the final flatten layer.

The flatten layer reshapes this input into a single dimension vector to feed into the MLP head layer with a GELU non-linearity activation function. In the proposed HViT4Lung model, the MLP Head consists of two dense layers in which the number of units in the second dense layer is half of the units in the first one which alleviates the connection to the final Classifier Head. The output of the MLP head feeds to the classifier head which is a dense layer with a softmax

Authorized licensed use limited to: University of Florida. Downloaded on October 30,2026 at 04:54:02 UTC from IEEE Xplore. Restrictions apply.

activation function. Each unit of the classifier head layer represents a separate class that works for the final results of the feature classification task. Therefore, the dimension of the final classifier unit is equal to the number of the classes N_{class}. Furthermore, with all MLP and MSA layers, the dropout layer is included to randomly set some of the input units to 0 at each step during training time, which helps prevent overfitting of the model.

For medical applications, there is usually a lack of enough labeled data. Unlike CNN models, transformers may not generalize well on small datasets due to the lack of inductive biases [9]. CNNs are equipped with locality, two-dimensional neighborhood structure, and translation equivariance. However, in the ViT, only MLP feed-forward layers are local while the self-attention layers are global. Nevertheless, the issue can be resolved if they are enhanced with transfer learning by being trained on large datasets. Therefore, the HViT4Lung model is aimed to be pre-trained on a larger dataset and then fine-tuned on the small target dataset. However, the Classifier Head needs to be modified in accordance with the number of classes in each dataset. Therefore, during fine-tuning, it should be replaced with the new Classifier Head and trained from scratch while the other layers are being fine-tuned.

IV. Experimental Results

In this section, the proposed HViT4Lung pipeline is being evaluated and validated on a real lung CT images dataset. Accordingly, a large dataset is used for pre-training of the proposed HViT4Lung model and then using transfer learning a real lung CT images dataset is applied for training, validation, and testing of the proposed pipeline which are described in the first subsection. Next, the implementation of the proposed HViT4Lung in Python is described and finally, the results are provided and discussed.

A. Dataset

In 2019, the Iraq-Oncology Teaching Hospital/National Center for Cancer Diseases (IQ-OTH/NCCD) collected lung CT scans captured from patients in different disease stages and healthy volunteers in DICOM format [35]. The dataset contains 1190 images representing the chest CT scans of 110 cases which differ in gender, age, occupation, location, and life-style. Lack of labeled data is a common issue in the area of medical imaging, however, one of the advantages of this dataset is that the images have been unidentified and labeled by oncologists and radiologists with three marks as "normal", "benign", and "malignant", as shown in Figure 2.

The scanner which is utilized to generate this dataset is the Siemens SOMATOM, with the protocols of slice thickness of 1 mm, window width range of 350 to 1200 Hounsfield Unit (HU), and a window center range of 50 to 600 HU with Deep Inspiration Breath Hold (DIBH). It is worth mentioning that due to the small size of IQ-OTH/NCCD dataset, the model is pre-trained using CIFAR-100 dataset [36] which consists of images of size 32×32 with 100 different classes.



(a) Normal case (b) Benign case (c) Malignant case

Fig. 2. Sample cases

B. HViT4Lung Model Implementation

The proposed HViT4Lung model has been implemented in Python using Keras [37] and TensorFlow [38] open-source libraries.[1] In this research, the model is built using Keras functional interface by leveraging the sequential model in which the layers can branch into multiple paths and have various inputs. Figure 3 shows the detailed architecture and input-output tensors size for the model. The implemented HViT4Lung model has more than 14 million parameters to train.

According to Fig. 3, the original input image may have the size of $(256 \times 256 \times 3)$. Then, the augmentation block with various layers shown in the structure, can resize the input image into any required dimension which is $(224 \times 224 \times 3)$ in our implementation. The new tensor is fed to the CNN block which consists of multiple convolution and average-pooling layers and the output has $(54 \times 54 \times 3)$ dimension. This serves as an input for the patching layer which the size of the patches is chosen to be $P_p = 6$. Considering the patch size and equation (3), the total number of patches obtained as 81. The patch encoder layer projects each patch into a vector of size $d_{proj} = 64$. Therefore, the output tensor is $(batch_{size} \times 81 \times 64)$ which is used as an input for the transformer block.

The implemented model has 12 transformer encoder blocks that are connected in series. Each block has two normalization layers, one MSA layer, two skip connections, and two dense layers (forming a feed-forward layer) as shown in Fig. 3. For the MSA layer, there are 6 transformer heads and the dropout rate is chosen to be 0.1. Then, a normalization layer followed by a flatten layer are included to reshape the output of the transformer block as $(5184 = 81 \times 64)$. This tensor is fed into the MLP head layer consisting of two dense layers with dimensions chosen to be 2048 and 1024 respectively which is connected to the final classifier layer. The final layer output has three dimensions representing three classes in our lung CT dataset. However, this layer was modified based on the number of classes for the pre-training dataset.

[1] A GitHub repository at https://github.com/Reza-rn/DL_HViT4Lung has been developed for HViT4Lung model, and access can be granted upon request.

Authorized licensed use limited to: University of Florida. Downloaded on October 30,2026 at 04:54:02 UTC from IEEE Xplore. Restrictions apply.

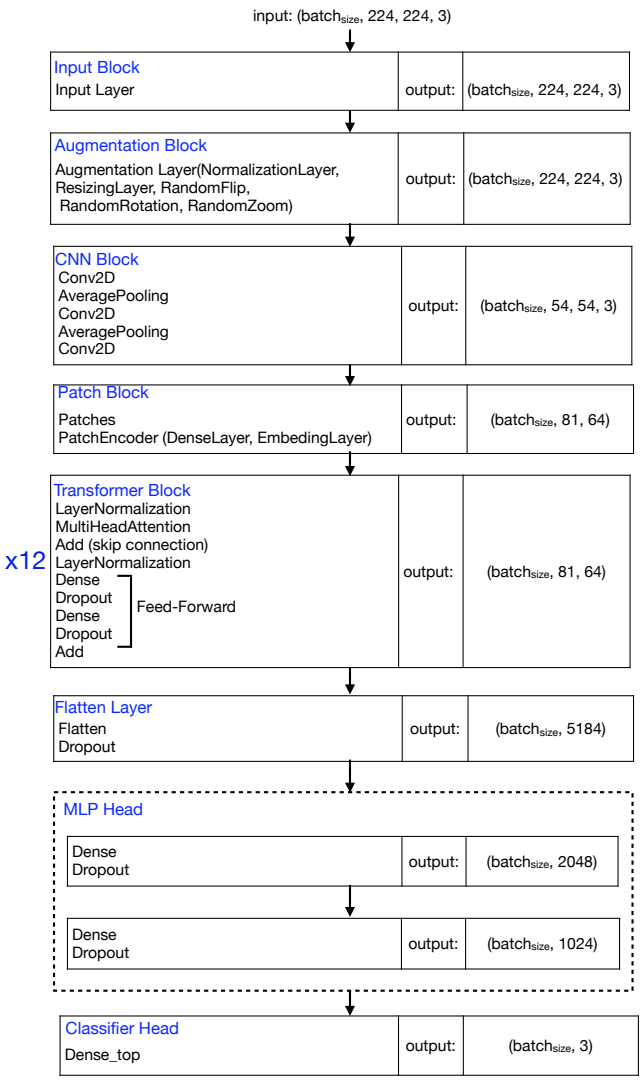
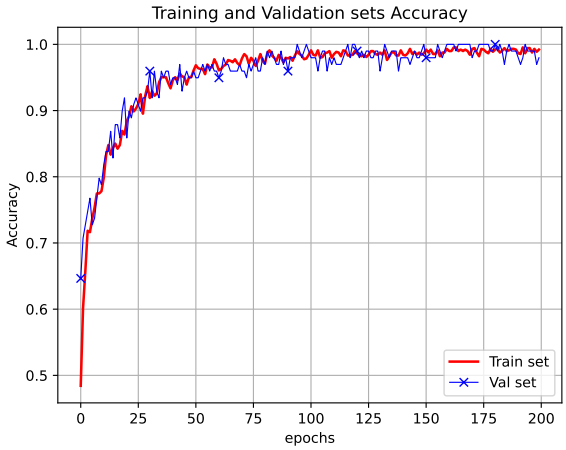
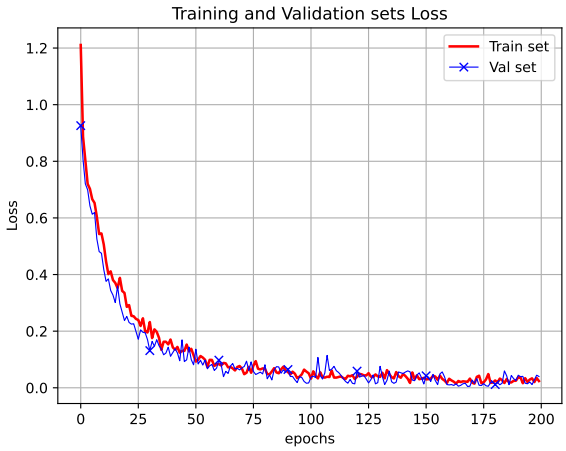


Fig. 3. HViT4Lung detailed implementation architecture



(a) Training and validation accuracy



(b) Training and validation loss

Fig. 4. Training and validation accuracy and loss for the proposed HViT4Lung model

C. Results

For the proposed model, the Adam optimizer with $learning_rate = 1e-3$ and $weight_{decay} = 1e-4$ is used to compile the model. Also, the size of the batches is chosen to be $batch_{size} = 256$. The proposed model is evaluated on a Linux PC with NVIDIA GeForce RTX 3070 for fast processing with CUDA 12.0, Python 3.9, and Tensorflow 2.11.

First, the model was pre-trained on the CIFAR-100 dataset with $\#epochs = 500$ while the classifier head is replaced with a dense layer including 100 units to represent each of the classes for the classification of this dataset. It is worth mentioning that the input images may have different dimensions, however, they are being resized to the target size by the augmentation layer.

Then, the pre-trained model is fine-tuned on the lung CT dataset by replacing the classifier head with the appropriate one and running the model for 200 epochs. The dataset is divided into the train, validation, and test sets with ratios of 80%, 10%, and 10%, respectively. The validation set is needed to show how well the model performs against known labeled cases while the test set evaluates the model against unseen cases. The obtained training accuracy using this model is 99.2% and the best validation accuracy is 99.09%. Figure 4 shows the accuracy and loss during training for both train and validation sets over 200 epochs. Accordingly, Fig. 4a shows the model achieves around 90% accuracy within 100 epochs, however, by running more epochs, the accuracy reaches very high values. Also, the loss function is defined to be the categorical cross entropy which the evolution of the value is shown in Fig. 4b for training and validation sets. It is obvious that the model loss is reaching very low values within the training period.

The related confusion matrix for the HVit4Lung model is calculated using scikit-learn module [39] as shown in Fig. 5. Then, based on those results, the model is evaluated using classification metrics such as precision, recall, F1-score, and support, which is demonstrated in Table I. According to this table, the overall accuracy of the model is 99.09% which

Authorized licensed use limited to: University of Florida. Downloaded on October 30,2026 at 04:54:02 UTC from IEEE Xplore. Restrictions apply.

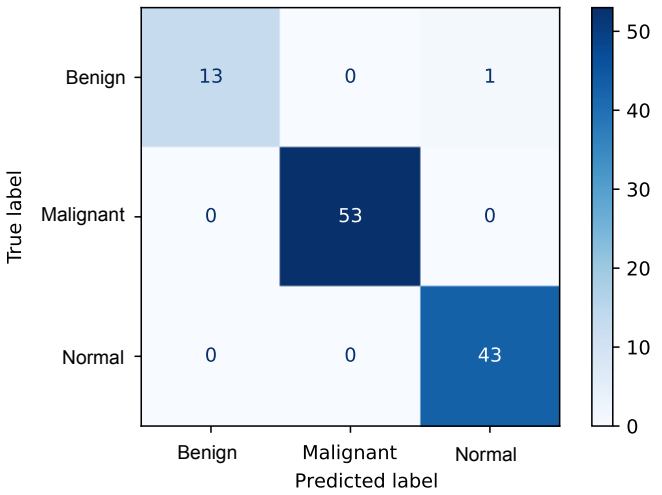


Fig. 5. Confusion matrix for the proposed HViT4Lung model

indicates how accurate and precise the proposed model is considering the total predicted cases. The overall weighted recall of 99.09% is obtained which demonstrates the ratio of the true positives to all actually predicted positives. Also, the weighted average F1-score of 99.08% is achieved which indicates the harmonic mean of recall and precision. The high F1-score of the proposed HViT4Lung model shows the high quality of the classifier and indicates how well the model works on the dataset. Also, the support metric per class is reported which indicates the number of images with true labels in each particular class of the test set.

TABLE I: Classification metrics

class	*precision*	*recall*	*F1-score*	*support*
Benign	1.0000	0.9286	0.9630	14
Malignant	1.0000	1.0000	1.0000	53
Normal	0.9773	1.0000	0.9885	43
accuracy			0.9909	110
macro avg	0.9924	0.9762	0.9838	110
weighted avg	0.9911	0.9909	0.9908	110

Meantime, we have tested various pre-trained CNN models with transfer learning including ResNet, Xception, MobileNet, Inception, and EfficientNet on the lung CT dataset by fine-tuning and modifying the top layer. Accordingly, the highest accuracy that was achieved among these models is about 96% for training and 89% for the test set which shows the superior performance of the proposed HViT4Lung pipeline over all those models.

V. Conclusion and Future Work

In this work, we proposed a novel lung CT scan classification method called HViT4Lung. The method is a deep-learning-based hybrid framework using vision transformers and CNNs, augmented by transfer learning that extracts features from the chest CT images to detect lung nodules and predict their malignancy. The main challenging part of this problem is to extract essential feature vectors from images. To address this problem, we utilized a hybrid vision transformer with CNN model by fine-tuning parameters in order to locate near-optimal parameter settings for our proposed model. With the proposed pipeline, the overall test accuracy of 99.09%, validation accuracy of 99.09%, and training accuracy of 99.20% are achieved, which is very encouraging and prominent compared to other strategies in the field. As future work, the model can be enhanced by adding parallel pipelines of the transformers and CNN blocks. Furthermore, this work can be expanded by preparing and labeling larger datasets of lung CT images. The proposed model can also be applied to classifying other medical imaging applications.

References

[1] R. L. Siegel, K. D. Miller, and A. Jemal, "Cancer statistics, 2019," *CA: A Cancer Journal for Clinicians*, vol. 69, no. 1, pp. 7–34, 2019. [Online]. Available: https://acsjournals.onlinelibrary.wiley.com/doi/abs/10.3322/caac.21551

[2] K. T. Chui, B. B. Gupta, H. R. Chi, V. Arya, W. Alhalabi, M. T. Ruiz, and C. W. Shen, "Transfer learning-based multi-scale denoising convolutional neural network for prostate cancer detection," *Cancers*, vol. 14, 8 2022.

[3] A. Del Ciello, P. Franchi, A. Contegiacomo, G. Cicchetti, L. Bonomo, and A. R. Larici, "Missed lung cancer: when, where, and why?" *Diagnostic and interventional radiology*, vol. 23, no. 2, p. 118, 2017.

[4] X. Shao, R. Niu, Z. Jiang, X. Shao, and Y. Wang, "Role of pet/ct in management of early lung adenocarcinoma," *American Journal of Roentgenology*, vol. 214, no. 2, pp. 437–445, 2020.

[5] X. Cui, M. A. Heuvelmans, S. Fan, D. Han, S. Zheng, Y. Du, Y. Zhao, G. Sidorenkov, H. J. Groen, M. D. Dorrius *et al.*, "A subsolid nodules imaging reporting system (ssn-irs) for classifying 3 subtypes of pulmonary adenocarcinoma," *Clinical lung cancer*, vol. 21, no. 4, pp. 314–325, 2020.

[6] A. Oikonomou, P. Salazar, Y. Zhang, D. M. Hwang, A. Petersen, A. A. Dmytriw, N. S. Paul, and E. T. Nguyen, "Histogram-based models on non-thin section chest ct predict invasiveness of primary lung adenocarcinoma subsolid nodules," *Scientific reports*, vol. 9, no. 1, p. 6009, 2019.

[7] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.

[8] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," in *Advances in Neural Information Processing Systems*, C. Cortes, N. Lawrence, D. Lee, M. Sugiyama, and R. Garnett, Eds., vol. 28. Curran Associates, Inc., 2015.

[9] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations*, 2021. [Online]. Available: https://openreview.net/forum?id=YicbFdNTTy

[10] T. Zhang, M. Yuan, Y. Zhong, Y.-D. Zhang, H. Li, J.-F. Wu, and T.-F. Yu, "Differentiation of focal organising pneumonia and peripheral adenocarcinoma in solid lung lesions using thin-section ct-based radiomics," *Clinical radiology*, vol. 74, no. 1, pp. 78–e23, 2019.

[11] C. Hassani, B. A. Varghese, J. Nieva, and V. Duddalwar, "Radiomics in pulmonary lesion imaging," *American Journal of Roentgenology*, vol. 212, no. 3, pp. 497–504, 2019.

[12] A. Bhandary, G. A. Prabhu, V. Rajinikanth, K. P. Thanaraj, S. C. Satapathy, D. E. Robbins, C. Shasky, Y.-D. Zhang, J. M. R. Tavares, and N. S. M. Raja, "Deep-learning framework to detect lung abnormality–a study with chest x-ray and lung ct scan images," *Pattern Recognition Letters*, vol. 129, pp. 271–278, 2020.

[13] W. Sun, B. Zheng, and W. Qian, "Computer aided lung cancer diagnosis with deep learning algorithms," in *Medical imaging 2016: computer-aided diagnosis*, vol. 9785. SPIE, 2016, pp. 241–248.

Authorized licensed use limited to: University of Florida. Downloaded on October 30,2026 at 04:54:02 UTC from IEEE Xplore. Restrictions apply.

[14] G. L. F. da Silva, T. L. A. Valente, A. C. Silva, A. C. de Paiva, and M. Gattass, "Convolutional neural network-based pso for lung nodule false positive reduction on ct images," *Computer methods and programs in biomedicine*, vol. 162, pp. 109–118, 2018.

[15] M. I. Faisal, S. Bashir, Z. S. Khan, and F. H. Khan, "An evaluation of machine learning classifiers and ensembles for early stage prediction of lung cancer," in *2018 3rd international conference on emerging trends in engineering, sciences and technology (ICEEST)*. IEEE, 2018, pp. 1–4.

[16] J. Hofmanninger, F. Prayer, J. Pan, S. Röhrich, H. Prosch, and G. Langs, "Automatic lung segmentation in routine imaging is primarily a data diversity problem, not a methodology problem," *European Radiology Experimental*, vol. 4, no. 1, pp. 1–13, 2020.

[17] Q. Dou, T. Y. So, M. Jiang, Q. Liu, V. Vardhanabhuti, G. Kaissis, Z. Li, W. Si, H. H. Lee, K. Yu *et al.*, "Federated deep learning for detecting covid-19 lung abnormalities in ct: a privacy-preserving multinational validation study," *NPJ digital medicine*, vol. 4, no. 1, p. 60, 2021.

[18] A. Saood and I. Hatem, "Covid-19 lung ct image segmentation using deep learning methods: U-net versus segnet," *BMC Medical Imaging*, vol. 21, no. 1, pp. 1–10, 2021.

[19] N. Paluru, A. Dayal, H. B. Jenssen, T. Sakinis, L. R. Cenkeramaddi, J. Prakash, and P. K. Yalavarthy, "Anam-net: Anamorphic depth embedding-based lightweight cnn for segmentation of anomalies in covid-19 chest ct images," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 3, pp. 932–946, 2021.

[20] Q. Dou, H. Chen, L. Yu, J. Qin, and P.-A. Heng, "Multilevel contextual 3-d cnns for false positive reduction in pulmonary nodule detection," *IEEE Transactions on Biomedical Engineering*, vol. 64, no. 7, pp. 1558–1567, 2016.

[21] H. Jung, B. Kim, I. Lee, J. Lee, and J. Kang, "Classification of lung nodules in ct scans using three-dimensional deep convolutional neural networks with a checkpoint ensemble method," *BMC medical imaging*, vol. 18, no. 1, pp. 1–10, 2018.

[22] Y. Gu, X. Lu, L. Yang, B. Zhang, D. Yu, Y. Zhao, L. Gao, L. Wu, and T. Zhou, "Automatic lung nodule detection using a 3d deep convolutional neural network combined with a multi-scale prediction strategy in chest cts," *Computers in biology and medicine*, vol. 103, pp. 220–231, 2018.

[23] J. Yang, H. Deng, X. Huang, B. Ni, and Y. Xu, "Relational learning between multiple pulmonary nodules via deep set attention transformers," in *2020 IEEE 17th international symposium on biomedical imaging (ISBI)*. IEEE, 2020, pp. 1875–1878.

[24] H. Zhang, Y. Peng, and Y. Guo, "Pulmonary nodules detection based on multi-scale attention networks," *Scientific Reports*, vol. 12, no. 1, p. 1466, 2022.

[25] E. Vieira, D. Ferreira, C. Neto, A. Abelha, and J. Machado, "Data mining approach to classify cases of lung cancer," in *Trends and Applications in Information Systems and Technologies: Volume 1*. Springer, 2021, pp. 511–521.

[26] W. Shen, M. Zhou, F. Yang, D. Yu, D. Dong, C. Yang, Y. Zang, and J. Tian, "Multi-crop convolutional neural networks for lung nodule malignancy suspiciousness classification," *Pattern Recognition*, vol. 61, pp. 663–673, 2017.

[27] M. Al-Shabi, K. Shak, and M. Tan, "Procan: Progressive growing channel attentive non-local network for lung nodule classification," *Pattern Recognition*, vol. 122, p. 108309, 2022.

[28] X. Liu, F. Hou, H. Qin, and A. Hao, "Multi-view multi-scale cnns for lung nodule type classification from ct images," *Pattern Recognition*, vol. 77, pp. 262–275, 2018.

[29] J. Yuan, X. Liu, F. Hou, H. Qin, and A. Hao, "Hybrid-feature-guided lung nodule type classification on ct images," *Computers & Graphics*, vol. 70, pp. 288–299, 2018.

[30] J. Pan, H. Zhang, W. Wu, Z. Gao, and W. Wu, "Multi-domain integrative swin transformer network for sparse-view tomographic reconstruction," *Patterns*, vol. 3, no. 6, p. 100498, 2022.

[31] Q. Lyu, S. V. Namjoshi, E. McTyre, U. Topaloglu, R. Barcus, M. D. Chan, C. K. Cramer, W. Debinski, M. N. Gurcan, G. J. Lesser *et al.*, "A transformer-based deep-learning approach for classifying brain metastases into primary organ sites using clinical whole-brain mri images," *Patterns*, vol. 3, no. 11, p. 100613, 2022.

[32] C. Niu and G. Wang, "Unsupervised contrastive learning based transformer for lung nodule detection," *Physics in Medicine & Biology*, vol. 67, no. 20, p. 204001, 2022.

[33] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. u. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., vol. 30. Curran Associates, Inc., 2017.

[34] Y. Lecun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.

[35] H. Alyasriy and A. Muayed, "The iq-othnccd lung cancer dataset," *Mendeley Data*, vol. 1, no. 1, pp. 1–13, 2020.

[36] A. Krizhevsky, G. Hinton *et al.*, "Learning multiple layers of features from tiny images," 2009.

[37] F. Chollet *et al.*, "Keras," https://keras.io, 2015, version 2.6.0.

[38] M. Abadi and et al., "TensorFlow: Large-scale machine learning on heterogeneous systems," 2015, software available from tensorflow.org. [Online]. Available: https://www.tensorflow.org/

[39] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg *et al.*, "Scikit-learn: Machine learning in python," *the Journal of machine Learning research*, vol. 12, pp. 2825–2830, 2011.

Authorized licensed use limited to: University of Florida. Downloaded on October 30,2026 at 04:54:02 UTC from IEEE Xplore. Restrictions apply.